

Topic: Performance Measures of Computer Architecture

Q. How is the performance of a computer system measured Discuss CPU Execution Time, Clock Rate, and the various factors that affect computer performance.

[DEF] Definition to Remember

Computer Performance measures how quickly a system executes tasks. It is inversely proportional to execution time ($\text{Performance} = 1 / (\text{Execution Time})$). The primary metric is CPU Execution Time, which depends on Instruction Count, CPI (Cycles Per Instruction), and Clock Rate.

CPU Performance Measures

The actual time the CPU spends computing a specific task (excludes I/O wait time).

2. Clock Rate and Clock Cycle

- * Clock Cycle (Period): The fundamental unit of time in a computer (e.g., 1 ns), generated by a hardware oscillator.

3. Other Key Performance Metrics

- * CPI (Cycles Per Instruction): Average number of clock cycles needed to execute one instruction.

4. Factors Affecting Computer Performance

Factor	Impact on Performance
Clock Speed	Higher clock speed executes instructions faster
Compiler Efficiency	Better compilers generate fewer machine instructions
Architecture / CPI	Pipelining, superscalar, and multi-core architectures improve CPI
Cache Memory	Fast cache reduces time the CPU spends waiting for data
Bus Width	A wider data bus transfers more data per cycle

[MUST-WRITE] Must-Write Points (for 10 marks)

- Performance is inversely proportional to execution time: $\text{Perf} = 1 / \text{Time}$.
- CPU Execution Time = (Instruction Count) x (CPI) x (Clock Cycle Time).
- Clock Cycle: Fundamental time unit; Clock Rate: Cycles per second (Hz).
- CPI: Cycles Per Instruction. Lower CPI means better architecture (via pipelining).
- MIPS: Millions of Instructions Per Second (measures execution speed rate).
- Compiler efficiency affects the total Instruction Count.
- Cache memory and Bus Width heavily impact performance by reducing data fetch delays.

[RECALL] Quick Recall

Performance = $1 / \text{Execution Time}$ -> Exec Time = (Instruction Count x CPI) / Clock Rate -> CPI = Cycles per Instruction -> Factors: Clock

Speed, Compiler, Pipelining (CPI), Cache, Bus Width